

Marvel by the Numbers: Visualizing Box Office Performance Across the MCU

Jose Lugo Gonzalez
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Regis University

This project looks at how Marvel Cinematic Universe movie performance has changed over time using box office data. The MCU is one of the most successful film franchises in modern entertainment, but its financial results have not been equal across every release year, individual movie, or release era. Because of that, this project focuses on the business side of the franchise rather than reviewing the movies themselves.

The main research question guiding this project is: How has Marvel Cinematic Universe movie performance changed over time based on worldwide box office revenue, production budget, and release-era performance? To answer this, I looked at revenue trends by year, the movies with the highest worldwide revenue, the relationship between production budget and worldwide revenue, and the average performance of different MCU release eras.

From a business standpoint, MCU success should be measured by more than total franchise revenue. A stronger analysis should look at when revenue peaked, which movies drove the largest results, and whether larger production budgets were consistently connected to stronger box office outcomes. The goal of this project is to use data visualization to explain how Marvel built box office momentum and where performance became less consistent after the franchise reached its peak.

Data

The dataset used for this project contains movie-level box office data for MCU films. Key variables used in the analysis included release date, production budget, opening weekend box office, domestic box office, and worldwide box office. These variables made it possible to compare the franchise from multiple angles instead of relying on one number alone.

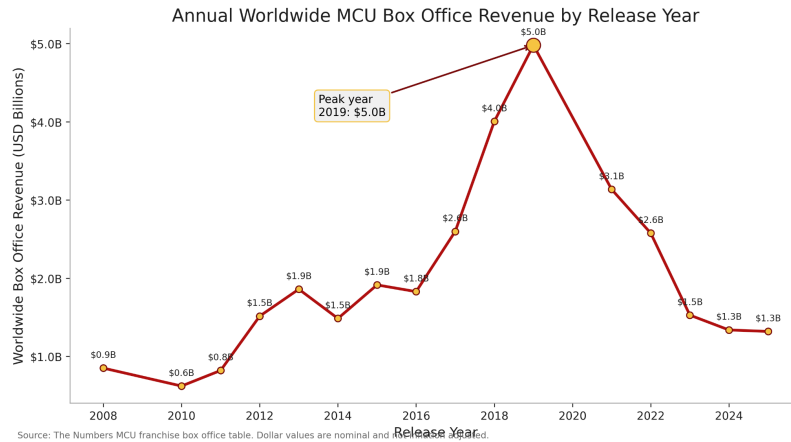
The most important outcome variable in this project was worldwide box office revenue. This variable was used to measure overall movie performance and to compare results by release year, individual movie, and release era. Production budget was also used to evaluate whether higher spending was generally connected to higher worldwide revenue.

Methodology

Python was used to clean, summarize, and visualize the data. Pandas was used to organize the movie-level dataset, create year-level and era-level summaries, and prepare values in millions or billions of U.S. dollars. Matplotlib was used to create the final visualizations. The charts were exported as PNG files and placed into both the slide presentation and this written paper.

Results

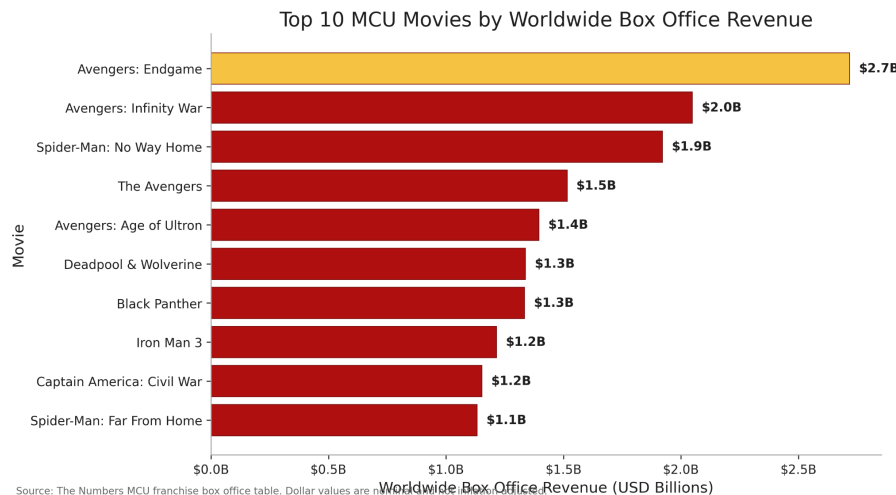
Figure 1: Annual Worldwide MCU Box Office Revenue by Release Year



The line chart shows how annual MCU worldwide box office revenue changed from the early foundation years through the more recent post-Endgame period. The chart shows a clear rise from the early years to the 2018 to 2019 peak period. The largest spike occurs around the Avengers-era releases, which suggests that major crossover films played a large role in driving the highest revenue years for the franchise.

After 2019, annual revenue became lower and more uneven. This does not mean the franchise stopped producing successful movies, but it does show that the post-Endgame period did not maintain the same level of yearly box office strength as the peak era.

Figure 2: Top 10 MCU Movies by Worldwide Box Office Revenue

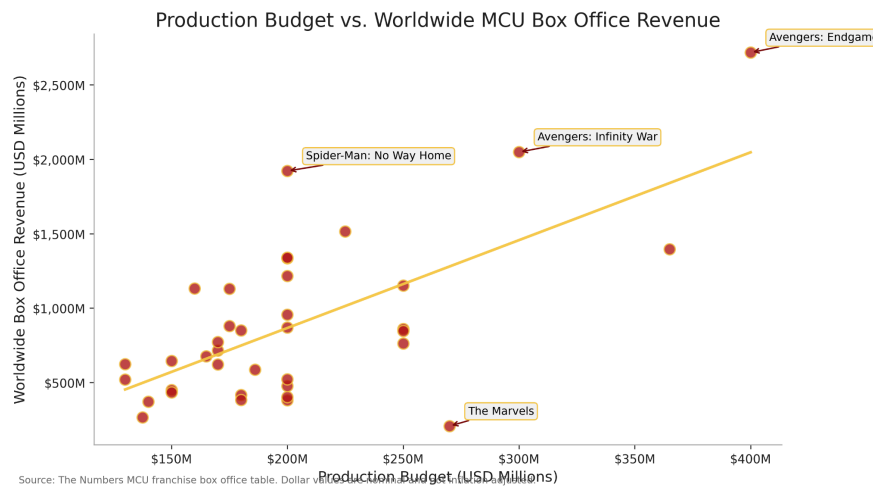


The horizontal bar chart compares the top 10 MCU movies by worldwide box office revenue. Avengers: Endgame stands out as the highest-grossing movie in the dataset at roughly \$2.7 billion worldwide. The rest of the top group is also dominated by Avengers films and major

crossover-style releases, which shows that the biggest MCU results were not evenly spread across all types of movies.

This visualization is useful because it shows revenue concentration. A small group of event films created a large share of the franchise's biggest box office outcomes. The sorted horizontal format also makes the movie titles easier to read and makes the ranking easier to compare.

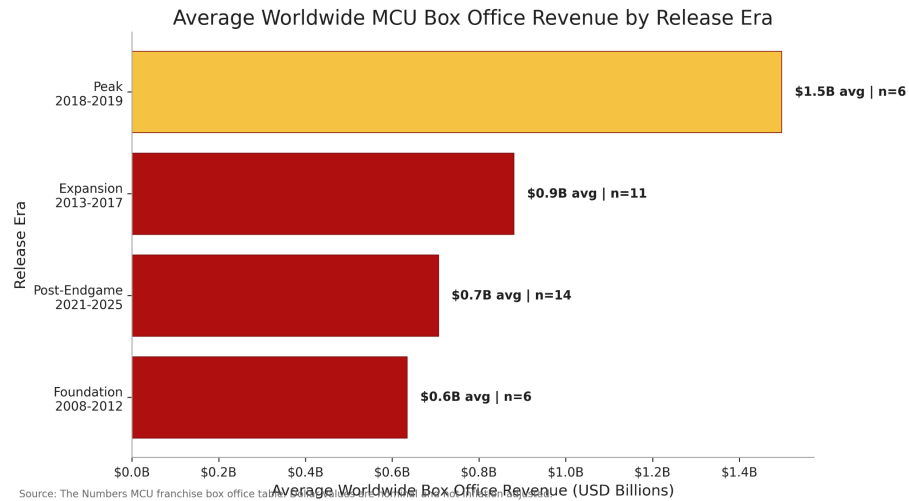
Figure 3: Production Budget vs. Worldwide MCU Box Office Revenue



The scatterplot compares production budget with worldwide box office revenue. There is a general positive relationship between budget and revenue, but the relationship is not perfect. Some movies earned much more than their budgets alone would suggest, especially major event films like Avengers: Endgame and Spider-Man: No Way Home.

The chart also shows that a larger budget does not automatically guarantee higher revenue. The Marvels is an example of a higher-budget film that did not generate revenue near the top performers. From a business perspective, spending matters, but audience demand, timing, brand momentum, and movie importance also matter.

Figure 4: Average Worldwide MCU Box Office Revenue by Release Era



The release-era bar chart summarizes the larger story by comparing average worldwide revenue across different MCU periods. The 2018 to 2019 peak era had the highest average revenue per movie. The expansion era performed well, while the post-Endgame era still produced successful films but had a lower average than the peak period.

This comparison shows that Marvel's strongest period was not only about releasing more movies. It was also about releasing movies with unusually high audience demand. The era-level view connects the individual charts into one larger story about franchise growth, peak performance, and more uneven results after the peak.

Conclusion

Overall, the data suggests that the Marvel Cinematic Universe built a powerful box office growth curve, but its performance was not evenly distributed. Revenue reached its strongest point during the 2018 to 2019 Avengers era, and a small group of major event films drove the highest worldwide totals. Production budget helped explain some results, but it did not fully explain success. The main takeaway is that Marvel's strongest outcomes came from the combination of budget, franchise timing, audience demand, and the importance of certain films within the larger MCU story.